Javed Ahmad

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Expertise & Research Focus

- **Core:** 3D computer vision | deep learning | multimodal scene understanding
- **3D Vision & Mapping:** SLAM | SfM | NeRF | 3D gaussian splatting | large-scale 3D reconstruction
- **Robotics:** robotic–arm motion planning | autonomous micro–UAV systems

Experience

09/2025 – Present **Computer Vision Researcher**, *DeepPlants, Rome*, Italy

- Designed and implemented an autonomous UAV-based perception pipeline on the ModalAI VOXL 2 platform, integrating PX4, MAVSDK/MAVLink, onboard mission control, telemetry, and real-time RGB streaming.
- Conducting aerial field-scanning missions with DJI Mavic 3 to acquire high-resolution imagery for SLAM benchmarking, 3D reconstruction, and agricultural scene understanding.

02/2024 – 08/2025 **Postdoctoral Researcher**, *Italian Institute of Technology (IIT)*, Italy

Advanced Robotics (ADVR), IIT Genova *02/2025 – 08/2025*

- Developed real-time SLAM pipelines using 3D Gaussian Splatting for immersive robotic teleoperation in the RIAP project.

Center for Cultural Heritage Technology (CCHT), IIT Venice *02/2024 – 01/2025*

- Built an autonomous 3D artifact-scanning system using UR5e/UR3e manipulators and an Artec3D scanner.
- Deployed the scanning system at CTE-Genova.
- Developed 6-DoF pose-tracking algorithms for complex cultural-heritage artifacts.

11/2020 – 12/2023 **Doctoral Researcher**, *PAVIS Lab, IIT*, Italy

- Developed a LiDAR–camera fusion framework to enhance 3D scene perception for autonomous-vehicle applications.
- Worked on multi-view 3D object localization for cultural-heritage augmented reality in the MEMEX Project.

07/2017 – 10/2020 **Research & Teaching Assistant**, *Information Technology University*, Pakistan

- Worked on 3D reconstruction and applied machine learning research.
- Served as a teaching assistant for graduate Machine Learning and undergraduate Calculus and Power Systems courses.

Technical Skills (including but not limited to)

Python	NumPy | SciPy | Pandas | Matplotlib | Multiprocessing | PyTest
Deep Learning	PyTorch | Transformers | DINO | U–Net | 2D/3D model training
3D Vision	OpenCV | Viser | Open3D | Nerfstudio | MMCV | NeRF | 3D Gaussian Splatting
Robotics SW	ROS 2 | Pinocchio | PyRoboplan | MoveIt 2 | UR–RTDE
Robotics Hardware	UR5e | UR3e | Raspberry Pi | RDK–X5 | Artec3D | Zivid | ZED | RealSense
3D Mapping	COLMAP | MASt3R/VGGT–SLAM | ViPE | DepthAnything 3 | MMDetection3D
Drones	ModalAI | PX4 | MAVSDK | MAVLink | VOXL 2 | Gazebo SITL
GPUs	NVIDIA V100/A100 | RTX A6000/4090/5090

Education

2020 – 2023 **Ph.D. in Computer Vision and Machine Learning**,
Italian Institute of Technology (IIT) & University of Genoa (UniGe), Italy
Thesis: *Multimodal 3D Scene Perception*

2017 – 2019 **Master of Science in Electrical Engineering**,
Information Technology University, Pakistan

2013 – 2017 **Bachelor of Science in Electrical Engineering**,
University of Central Punjab, Pakistan

Selected Publications

arXiv 2025 **Ahmad, J.**, et al. *From Fields to Splats: A Cross-Domain Survey of Real-Time Neural Scene Representations*. arXiv:2509.23555

arXiv 2025 **Ahmad, J.**, et al. *Hands-Free Heritage: Automated 3D Scanning for Cultural Heritage Digitization*. arXiv:2510.04781

IEEE/ASME MESA 2024 **Ahmad, J.**, et al. *Automated Artifacts Position and Orientation Estimation in Cultural Heritage*. 2024 IEEE/ASME International Conference on Mechatronic and Embedded Systems and Applications (MESA). [Teaser] [Presentation] [IEEE Xplore DOI]

arXiv 2023 **Ahmad, J.**, Del Bue, A. *mmFUSION: Multi-modal Fusion for 3D Object Detection*. [arXiv Link] [Presentation]

ICIAP 2022 **Ahmad, J.**, Toso, M., Taiana, M., James, S., Del Bue, A. *Multi-view 3D Object Localization from Street-level Scenes*. International Conference on Image Analysis and Processing (ICIAP), 2022. [Springer DOI] [GitHub Code] [Presentation]

Machine Vision and Applications 2022 Castro, E., ..., **Ahmad, J.**, et al. *Fill in the Blank for Fashion Complementary Outfit Product Retrieval*. Journal of Machine Vision and Applications, 2022. [DOI: 10.1007/s00138-022-01359-x]

Awards & Prizes

1st Prize Winner at VISUM 2021 and EEML 2020

Merit-based Master's Fellowship, ITU Pakistan

Reviewer Activity

IEEE RA-L, ICRA 2025, and IROS 2025

Summer Schools

2023 **ICVSS, Sicily**, *poster on 3D scene perception* [Poster]

2022 **Vision and Sports, Prague**

2021 **VISUM, Porto**, *competition winner* [Project]

2020 **EEML, Poland**, *proposal winner* [Proposal]

2020 **OXML, London**, fully sponsored based on academic profile

Community Service

Former member of Akhuwat Foundation, Pakistan, contributed to charity fundraising initiatives.